

Question	Answer	Marks
2(a)(i)	(time / $t =$) $9(.0) / 9.5 = 0.95$ (s)	A1
2(a)(ii)	$(u_V) = 9.5 \tan 38^\circ$ or $9.5 / \tan 52^\circ$ or $9.5 = u \cos 38^\circ$ and $u_V = u \sin 38^\circ$ or $9.5 = u \cos 38^\circ$ and $u_V = (u^2 - 9.5^2)^{1/2}$	C1
	$u_V = 7.4 \text{ m s}^{-1}$	A1
2(a)(iii)	$s = ut + \frac{1}{2}at^2$ $(h =) 7.4 \times 0.95 - \frac{1}{2} \times 9.81 \times 0.95^2$	C1
	$h = 2.6 \text{ m}$	A1
2(b)	(collision is) inelastic	B1

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2(c)	resultant force (in any direction) is zero	D1